

Ans: On Page 13, just above Theorem 1, it is mentioned that any nonzero matrix may be row reduced (that is, transformed by elementary row operations) into more than one matrix in echelon form, using different sequences of row operations.

∴ The statement is False.

b) The pivot positions in a matrix depend on whether row interchanges are used in the row reduction process.

Ans: The statement is False. By definition, a pivot position in a matrix A is a location in A that corresponds to a leading 1 in the reduced echelon form of A . By Theorem 1, the rref form of a matrix is unique. ∴ The pivot positions of a matrix are unique.

c) Reducing a matrix to echelon form is called the forward phase of the row reduction process.

Ans: On Page 15, just under the section "The Row Reduction Algorithm" it is mentioned that "The algorithm that follows consists of four steps, and it produces a matrix in echelon form." On Page 17, just above the numerical note it is mentioned that "The combination of steps 1-4 is called the forward phase of the row reduction algorithm."

d) Whenever a system has free variables, the solution set contains many solutions. Combining this two statements, we can see why the given statement is True.

e) A general solution of a system is an explicit description of all solutions of the system.

Ans: This statement is True. For reference, read Example 4.